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Roll No.....

End Semester Examination

Name of the Program: B.Tech(ECE) Semester: IV

Name of the Course: Communication Systems I Course Code: TEC 401

Time: 3 Hours Maximum Marks: 100

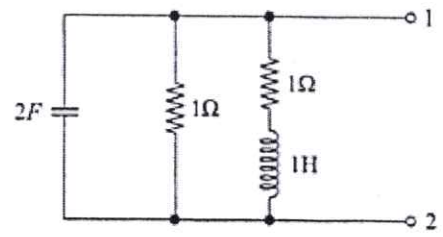
Note:

(i) All questions are compulsory.

(ii) Answer any two sub questions among a, b & c in each main question

(iii) Total marks in each main question are twenty

(iv) Each question carries 10 marks.

Q1	(20 marks)	
(a)	Explain the operation of the Ring modulator circuit which generate the DSB-SC wave.	**CO1**
(b)	(i) Show that a signal and its Hilbert transform are orthogonal. (ii) Derive the Power and Current Relations for Single-tone amplitude modulated wave.	
(c)	Explain the modulation and demodulation of VSB wave using filter method with the help of neat block diagram.	
Q2	(20 marks)	
(a)	An angle modulated signal with carrier frequency $\omega_c = 2\pi \times 10^5$ is described by the equation $x(t) = 10\cos(\omega_c t + 5\sin 3000\pi t + 10\sin 2000\pi t)$ (i) Find the power of the modulated signal. (ii) Find the frequency deviation. (iii) Find the deviation ratio. (iv) Find the phase deviation. (v) Estimate the bandwidth of $x(t)$.	**CO 2**
(b)	Explain with necessary mathematical relations, the demodulation of an FM wave using PLL.	
(c)	Derive an expression for a single tone frequency modulated wave.	
Q3	(20 marks)	
(a)	Find the power spectral density of the thermal noise voltage across terminal 1, 2 of a Passive network shown in figure below.  	**CO3**
(b)	Derive the expression for noise equivalent bandwidth and establish the relation between noise equivalent bandwidth and 3-dB bandwidth for a RC lowpass filter.	

(c)	Derive an expression for the overall equivalent noise figure for cascaded stages of amplifiers.	
Q4	(20 marks)	**CO4**
(a)	Show that in a PM system the figure of merit, assuming sinusoidal modulation, is given by $\gamma_{PM} = \frac{1}{2} m_p^2$, where m_p is the modulation index for PM.	
(b)	What is pre-emphasis? Why is it used? Sketch a typical pre-emphasis circuit and explain why de-emphasis must be used also.	
(c)	Assuming sinusoidal modulation, show that in an AM system with envelope detection the figure of merit is given by $\gamma_{AM} = \frac{\mu^2}{2+\mu^2}$, where μ is the modulation index for AM.	
Q5	(20 marks)	**CO5 & CO6**
(a)	With the help of neat circuit diagram, explain the generation and demodulation of PWM.	
(b)	State and prove sampling theorem.	
(c)	Draw the block diagram of communication system and explain the function of each block.	

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End Semester Examination JUNE 2024

Name of the Course: ECE **Semester: IVth**
Name of the Paper: Microprocessor and Its Applications **Paper Code: TEC 403**

Time: 3 Hours **Maximum Marks: 100**

Note :-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20marks)	CO3 & CO4
(a)	Explain memory organization in 8086.	
(b)	Explain the difference between following instructions i) ROL and RCL ii) AAM and AAD iii) NOT and NEG	
(c)	Write 8086 assembly language program to transfer 25 bytes of data from location 2050 to 3050 using string instructions.	
Q2	(20marks)	CO1 & CO2
(a)	Draw Timing diagram of ANI 32.	
(b)	Write a program to generate a delay of 2ms in 8085.	
(c)	Write a 8085 assembly language program to convert a Binary number to its GRAY code equivalent.	
Q3	(20marks)	CO3 & CO4
(a)	Explain Control word register of 8254. What are the different modes of 8254?	
(b)	Write a program in 8086 to find cube of an 8-bit number.	
(c)	Explain the functions of 8086 maximum mode pins.	
Q4	(20marks)	CO4
(a)	Draw Block diagram of Intel 8086 Microprocessor. Explain concept of Pipelining?	
(b)	Write 8086 assembly language program to convert a BCD number to its Binary equivalent.	
(c)	Write addressing mode of following instructions also explain the operation i) ADD [4500], BX ii) MOV SI, 3030 iii) DAS iv) XOR AX, [BX]	
Q5	(20marks)	CO5
(a)	Explain block Diagram of 8259.	

(b)	Program 8255 for the following configurations: To get 8-bit data from Port A multiply with content of port B and display result at Port C.	**& CO6**
(c)	Explain working of 8255 in BSR mode.	

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Roll No. TEC-404

B. Tech End Semester Examination 2024

IV Semester

Electromagnetic Field Theory

Time: Three Hours MM: 100

Note:

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks for each main question is twenty.
(iv) Each question carries 10 marks.

Q1	(20 marks)	**CO1**
(a)	For the given vectors $\vec{A} = 2\hat{a}_x - 3\hat{a}_y + 5\hat{a}_z$, $\vec{B} = 4\hat{a}_x + 3\hat{a}_y - 2\hat{a}_z$ and $\vec{C} = \hat{a}_x + \hat{a}_y + \hat{a}_z$ determine (i) vector component of $\vec{A}$ along $\vec{C}$, (ii) a unit vector perpendicular to both $\vec{A}$ and $\vec{B}$, (iii) vector triple product of $\vec{A}$, $\vec{B}$ and $\vec{C}$.	
(b)	Express the following vectors is Cartesian coordinates system and determine its value at (2, 45°, 5). $\vec{A} = \rho(z^2+1)\,\hat{a}_\rho - \rho z \cos\phi\ \hat{a}_\phi$	
(c)	Determine the divergence and curl of the following vector field and evaluate them at the specified points. $\vec{B} = \rho^2 \cos\phi\ \hat{a}_\rho + 3\rho z \cos\phi\ \hat{a}_\phi$ at (7, π/2, 2)	
Q2	(20 marks)	**CO2**
(a)	In free space, $V = x^2 y(z+3)$ V. Find (i) E at (3, 4, -6) (ii) the charge within the cube $0 < x < 1, 0 < y < 1, 0 < z < 1$.	
(b)	Show that the tangential component of magnetic flux strength and normal component of magnetic flux density is continuous across the boundary, i.e., $\vec{H}_{t1} = \vec{H}_{t2}$ and $\vec{B}_{n1} = \vec{B}_{n2}$. Where symbols have their usual meaning. 1 represents the medium 1 and 2 represent the medium 2.	
(c)	A homogeneous dielectric ($\varepsilon_r = 2.5$) fills region 1 ($x < 0$) while region ($x > 0$) is free space. If $\vec{D}_1 = -12\hat{a}_x - 10\hat{a}_y + 4\hat{a}_z$ nC/m², find $\vec{D}_2$ and θ_2 (angle between $\vec{D}_2$ and normal to interface).	
Q3	(20 marks)	**CO3**
(a)	Write the Maxwell's equation in both differential and integral form for the following cases (i) Instantaneous (or general) form (ii) Time harmonic form (iii) for static field (iv) for charge free region.	**CO6**

(b)	The magnetic field component of an EM wave propagating through a nonmagnetic medium is $\vec{H} = 25\sin(2\times10^8 t + 6x)\hat{a}_y$ mA/m. Determine (a) The direction of wave propagation (b) The permittivity of the medium (c) The electric field intensity (d) wave impedance (e) Displacement current density.	
(c)	In a nonmagnetic material $\vec{H} = 30\cos(2\pi\times10^8 t - 6x)\,\hat{a}_y$ mA/m. (a) Find the intrinsic impedance, (b) the Poynting vector and (c) the time average power crossing the surface $x = 1$, $0 < y < 2$, $0 < z < 3$ m.	
Q4	(20 marks)	CO4
(a)	Show that a Quarter Wave Transformer can be used for impedance matching between load and transmission line.	CO6
(b)	A distortionless line operating at 120 MHz has R = 20 Ω/m, L = 0.3 μH/m, and C = 63 pF/m. (a) Determine γ, u and Zo (b) How far will a voltage wave travel before it reduces to 20% of its initial value (c) How far will it travel to suffer a 45° phase shift?	
(c)	A distortionless line has Zo = 60 Ω, α = 20 mNp/m, u = 0.6c, where c is the speed of light in a vacuum. Find R, L, G, C and λ at 100 MHz.	
Q5	(20 marks)	CO5
(a)	What do you understand by TE, TM and TEM mode of parallel plate waveguide? Draw the field lines for TEM, TE_1, and TM_1 for parallel plate waveguide.	
(b)	Maxwell carried out the modification in Ampere's law to make it consistent with continuity equation. What was that modification? Determine the Maxwell's fourth equation based on that modification.	
(c)	Define the term wave polarization. Write the most general form of the Electric field equation and then provide the condition of linear, circular and elliptical polarization. Determine the state of polarization for following E-filed. If circular or elliptical also determine the sense of rotation. $\vec{E} = j5e^{-3z}\hat{a}_x + 3e^{-3z}\hat{a}_y$	